MOTHERBOARD & CHIPSET

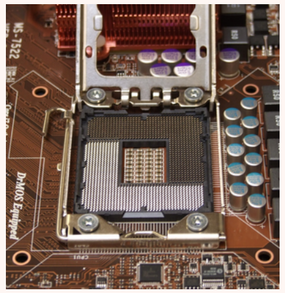
Alejandro Cáliz Royo

WHAT IS A MOTHERBOARD?

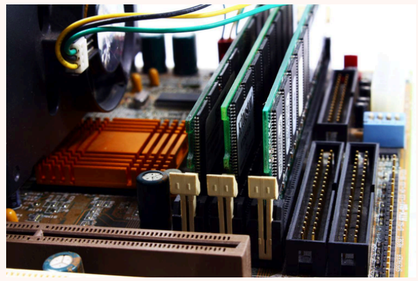
Is the main printed circuit board in genral-purpose computers and other expandebles systems. It holds and allows communications between many of the crucial electronic components of a system, such as the central processing unit and memory, and provides connectors for others peripherals.



PARTS OF A MOTHERBOARD



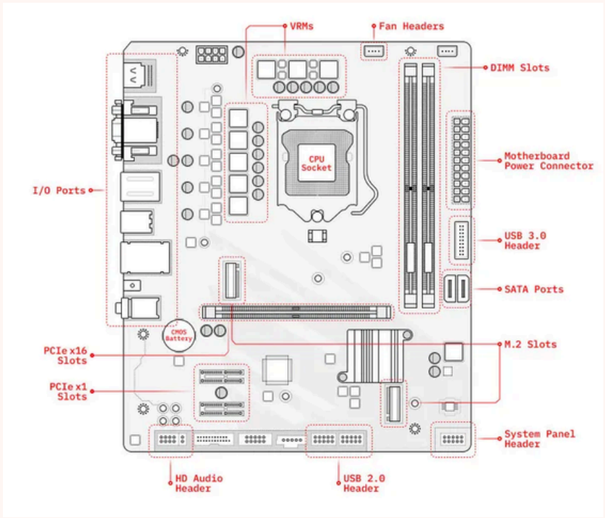
CPU Socket



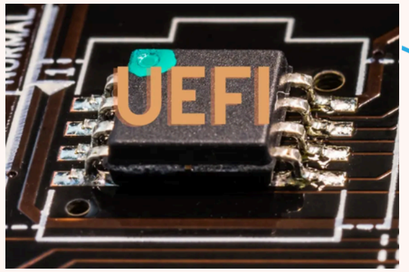
Memory Slots

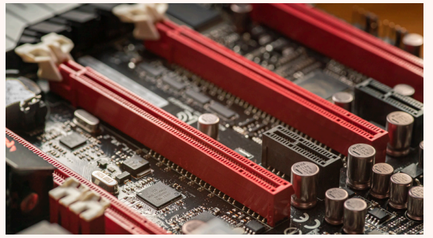


BIOS



UEFI





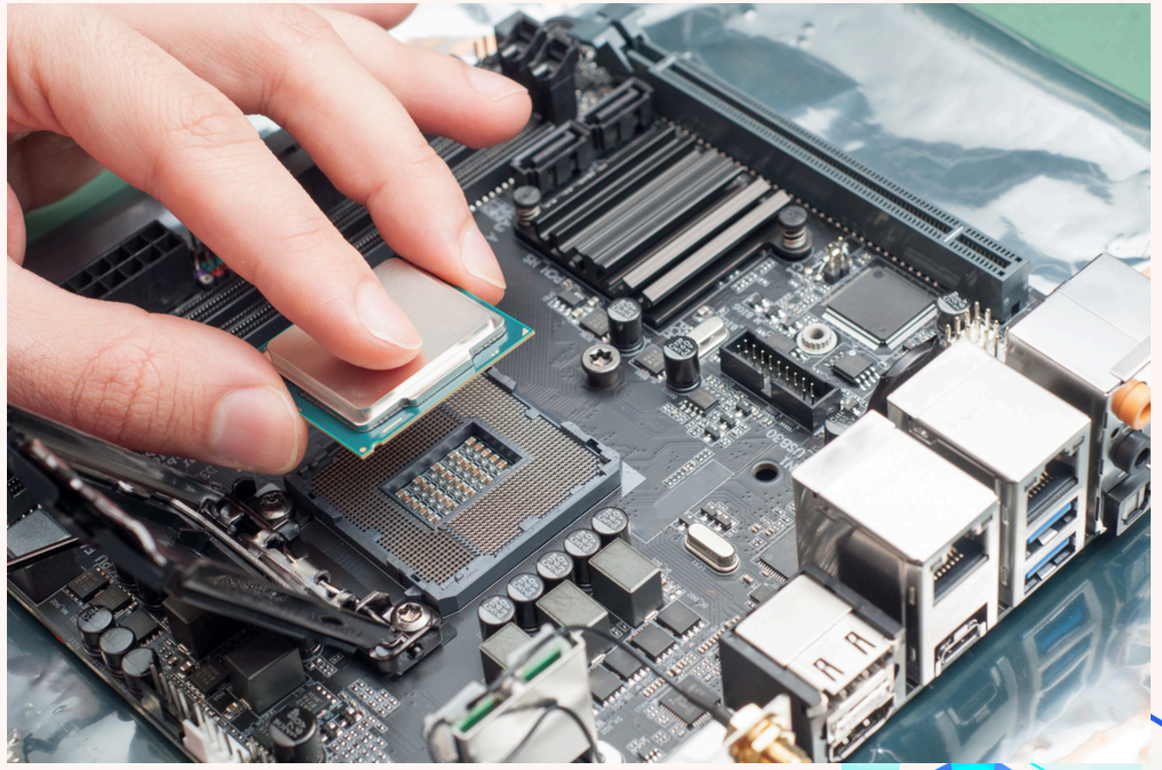
Texto

Texto



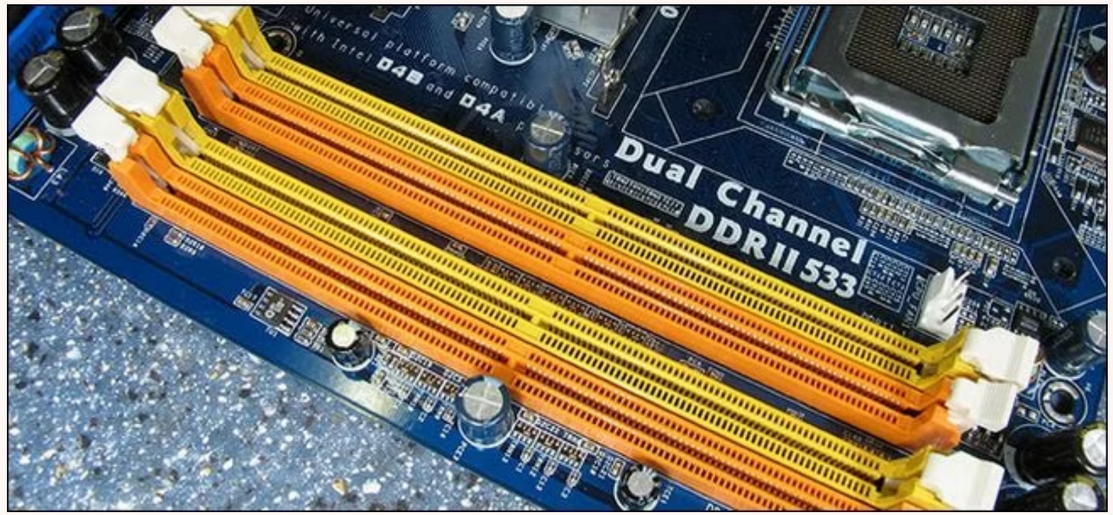
CPU SOCKETS

CPU sockets contain one or more mechanical components providing mechanical and electrical connections between a microprocessor and a printed circuit board (PCB). This allows for placing and replacing the central processing unit (CPU) without soldering.



MEMORY SLOTS

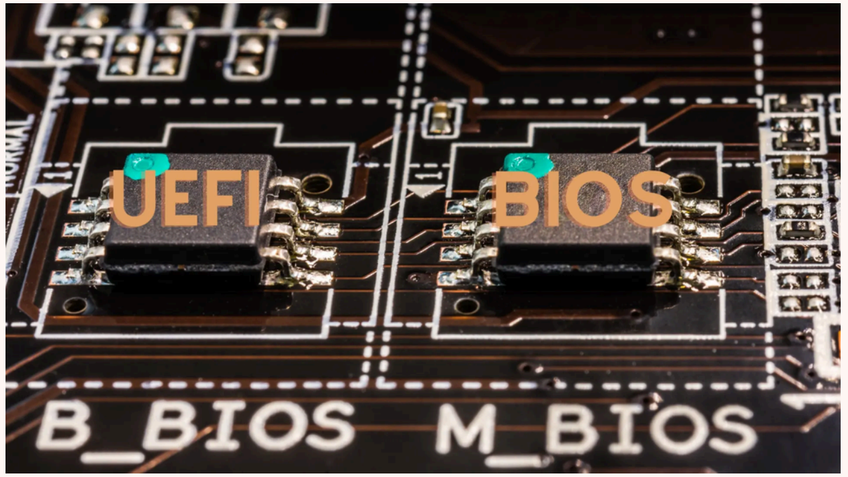
Memory slots into which the system's main memory is to be installed, typically in the form of DIMM modules containing RAM chips. Some motherboards have dual memory slots which allows the installation of different memory types, such as either DDR2 or DDR.



BIOS AND UEFI

BIOS

- Tests hardware components like the CPU, RAM, and storage drives to ensure they work properly before booting.
- Wakes up and configures the computer's internal hardware and input devices.
- Finds the operating system on your hard drive, SSD, or USB and hands over control to it.
- Allows you to change system settings like boot order, CPU/RAM overclocking, and hardware monitoring through a setup menu.



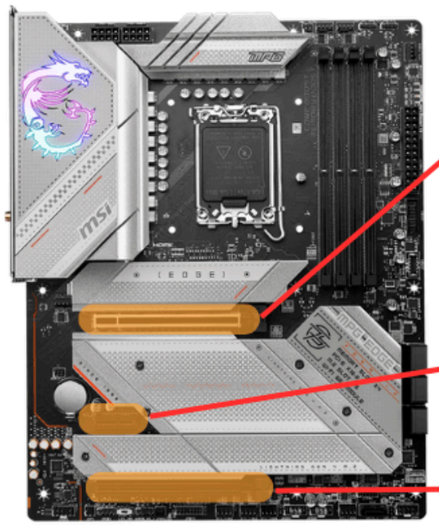
UEFI

UEFI is most modern computers used instead of a traditional legacy BIOS. UEFI is faster, supports larger storage drives (over 2TB), and includes modern security features like Secure Boot, though people still commonly use "BIOS" as a general term for both.



PCIE

PCIe (Peripheral Component Interconnect Express) is a high-speed motherboard interface standard used to connect hardware components, such as graphics cards, storage drives, and network adapters, directly to the CPU and system chipset



PCIe x16 Slot

Recommended Installation: Graphics Card

PCIe x1 Slot

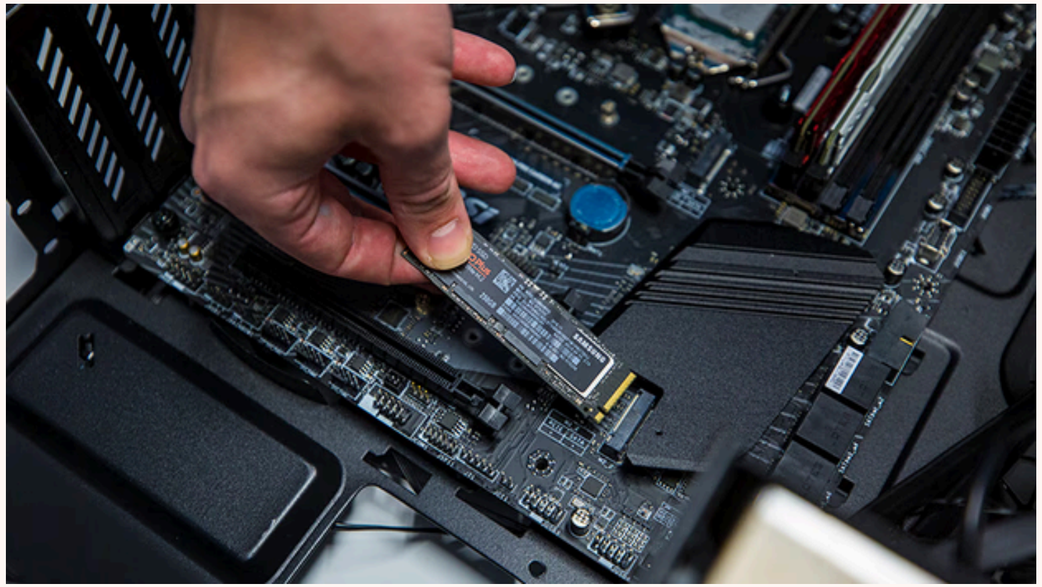
Recommended Installation: Network Card

PCIe x16 Slot

Recommended Installation:
SSD Expansion Card, Capture Card

- PCIe x16 Slots: These are the longest slots on the motherboard. They are located closest to the CPU and are primarily used for graphics cards (GPUs).
- PCIe x1 and x4 Slots: These are much shorter slots positioned lower down on the board. Most used on Wi-Fi cards, sound cards, capture cards, or network adapters go into these smaller slots.

M.2 PORT



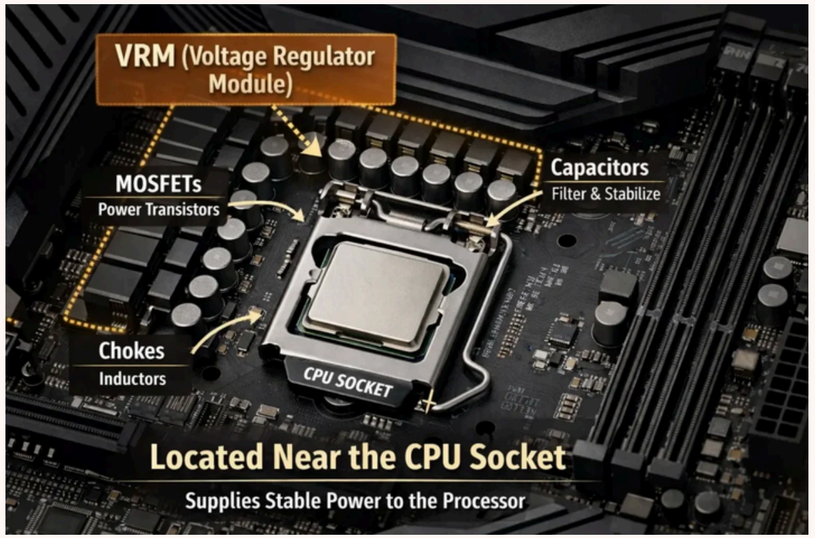
An M.2 port is a small, flat internal connector on a motherboard used to plug in compact expansion cards. An M.2 port can run on different underlying technologies:

- PCIe / NVMe: The modern standard for fast performance, connecting directly to the CPU or chipset to deliver high data transfer speeds.
- SATA: An older, slower type of M.2 drive that caps out at traditional SATA speeds and often disables a standard SATA port on the motherboard if used.

VRM

VRM (Voltage Regulator Module) is an electrical circuit that converts the 12-volt power from your power supply into the lower, stable voltage required by your CPU.
What the VRM does:

- Steps down voltage
- Cleans and stabilizes
- Handles power loads

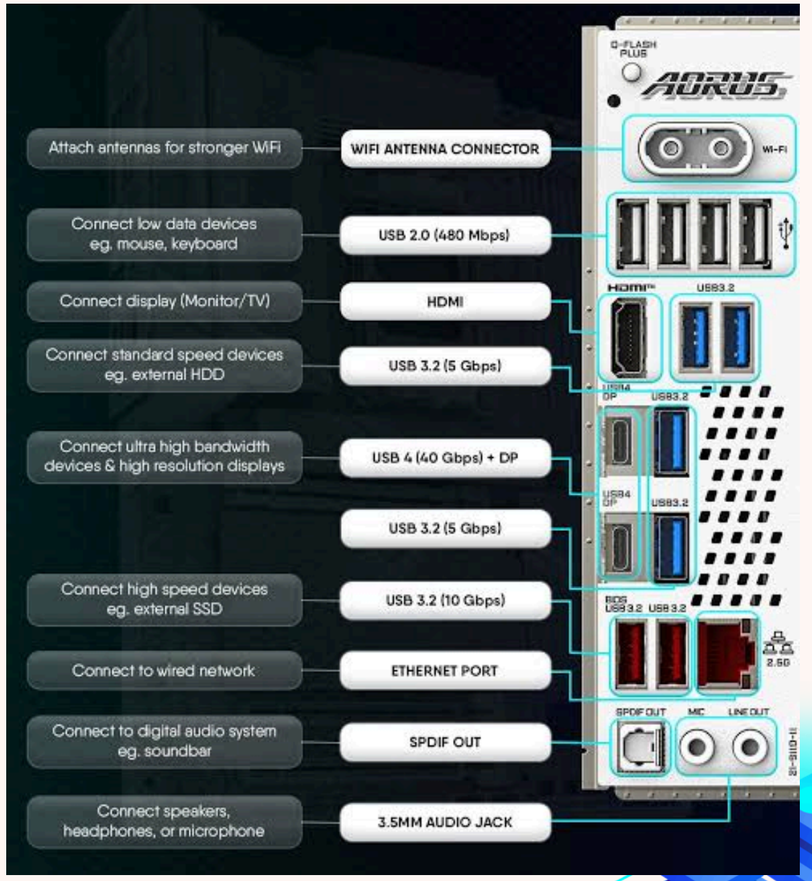


THE REAR I/O

The rear I/O (input/output) panel is the collection of physical ports on the back of a computer motherboard, exposed through the rear of the PC case.

Common Rear I/O Ports:

- USB Ports.
- Video Outputs like HDMI.
- Network / Ethernet.
- Wi-Fi Antenna Connectors.
- Audio Jacks & S/PDIF.
- Specialty Buttons.
- Legacy Ports.

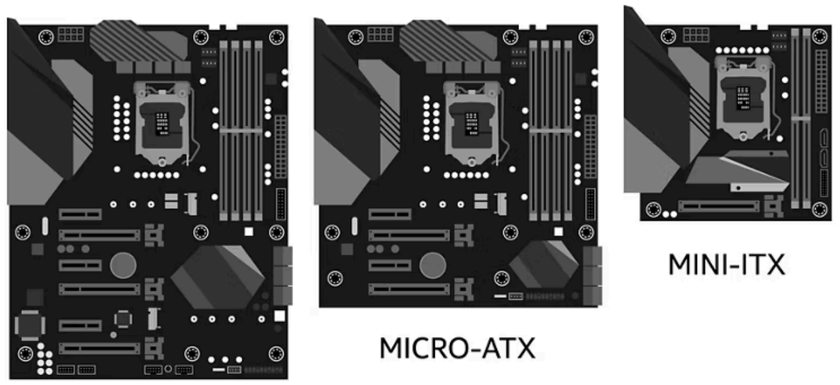


FORM FACTOR

Micro-ATX (244 x 244 mm): A mid-sized board with up to 4 RAM slots and 2 to 3 PCIe slots. It fits in both Micro-ATX and ATX cases.

ATX (305 x 244 mm): The standard full-size board. It includes 4 RAM slots and 3 to 5 PCIe slots, fitting mid-tower or full-tower cases.

Mini-ITX (170 x 170 mm): The smallest common size, featuring 2 RAM slots and 1 PCIe slot. Built for tiny, space-saving desktop cases.



ATX

MICRO-ATX

MINI-ITX



CHIP AND CHIPSET

What is the difference between a chip and a chipset?
A chip is a single piece of silicon that performs calculations or handles specific processing tasks. A chipset is a group of chips that manages data flow between the processor, memory and peripheral devices.



Chip

Chipset



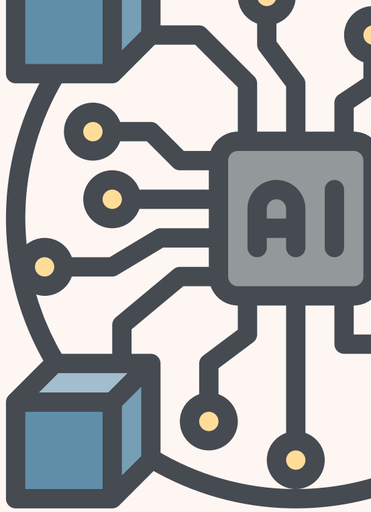
WHAT CHIPSET DOES

Chipset acts as a communication hub. It manages data traffic between the CPU and peripheral devices like USB ports, SATA drives, audio, network cards, and secondary PCIe lanes. While the CPU handles memory, graphics, and primary storage directly, the chipset expands those capabilities into the broader array of ports and connections found on the motherboard.





HOW CHIPSETS DIFFER WITHIN A PLATFORM



Entry-Level / Budget Tiers (e.g., Intel H-series/H810, AMD A-series/A620)

- Purpose: Basic office work, media streaming, or simple single-GPU setups.
- Features: Limited USB ports, fewer PCIe lanes, slower or fewer high-speed storage slots, and no support for CPU overclocking

Mainstream / Mid-Range Tiers (e.g., Intel B-series/B860, AMD B-series/B650/B850)

- Purpose: Best value for standard gaming and general-purpose custom PCs.
- Features: Balanced connectivity, support for memory/CPU overclocking on most mid-range lines, and a healthy mix of modern PCIe and USB connections.

Enthusiast / High-End Tiers (e.g., Intel Z-series/Z890, AMD X-series/X870, X870E)

- Purpose: Extreme gaming rigs, heavy content creation, and multi-drive storage arrays.
- Features: Maximum PCIe 5.0 lane allocation, extensive USB/Thunderbolt/USB4 support, robust power delivery support for overclocking, and extra "E" variants that double connectivity potential.

EXAMPLES



ASUS X870E ROG Crosshair Dark Hero AMD AM5 ATX Motherboard

SKU 994384
REMOVE ITEM
$638.99
ADD TO CART



Gigabyte B850M EAGLE WIFI7 AMD AM5 microATX Motherboard

SKU 033597
REMOVE ITEM
$169.99
ADD TO CART

General Specifications		
Packaging	Retail	Retail
Chipset		
Chipset	AMD X870E; AMD X870E	AMD B850; AMD B850
BIOS	64Mb AMI UEFI Bios	256 Mb Flash ROM, UEFI AMI BIOS
Manageability	WOL	PnP 1.0a WfM 2.0 SM BIOS 2.7 ACPI 5.0 DMI 2.7
CPU Support		
AMD CPU Support	Ryzen	Ryzen
Socket Type	AM5	AM5
Memory Specifications		
ECC	Non-ECC; Non-Registered ECC; Non-Registered	Non-ECC; Non-Registered
Memory Type	DDR5-4800 DDR5-5200 DDR5-5400 DDR5-5600 DDR5-5000 DDR5-5800	DDR5-5600 DDR5-6400 DDR5-6200 DDR5-6000 DDR5-5800
Memory Speeds Supported	8200+ MHz (OC)	8200(O.C) / 8000(O.C) / 7950(O.C) / 7900(O.C) / 7800(O.C) / 7600(O.C.) / 7400(O.C.) / 7200(O.C.) / 7000(O.C.) / 6800(O.C.) / 6600(O.C.) / 6400 / 6200 / 6000 / 5800 / 5600 MT/s
Memory Slots	4 x 288pin DIMM	4 x 288pin DIMM
Memory Channel Support	Dual Channel	Dual Channel
Max Memory Supported	256GB	256GB
Maximum Memory Supported Per Slot	64GB	64GB
Video Specifications		
Onboard Graphic Chipset	Requires CPU with Integrated Graphics	Requires CPU with Integrated Graphics
Storage Specifications		
SATA Express	5 x M.2 Socket 3 M key	2 x M.2 Socket 3 M Key
M.2 Port Type	2242 2260 2280 22110	2280 22110 25110 2580
SATA 6Gb/s	4	4
RAID Support	RAID 0 RAID 1 RAID 5 RAID 10	RAID 0 RAID 1 RAID 5 RAID 10

Audio Specifications		
Audio Chipset	ROG SupremeFX ALC4082	N/A
Audio Channels	7.1	7.1
Networking		
WiFi Support	Onboard WiFi	Onboard WiFi
LAN Chipset	Realtek 10Gb Ethernet	Realtek 2.5Gb Ethernet
LAN 2 Chipset	Realtek 5Gb Ethernet	N/A
LAN Speed	100/1000/2500/5000/10000Mbps	10/100/1000/2500Mbps
Wireless LAN Chipset	MediaTek Wi-Fi 7 MT7927	Realtek Wi-Fi 7 RTL8922AE
Wireless Technology	802.11b 802.11g 802.11a WiFi 4 (802.11n) WiFi 5 (802.11ac) WiFi 6 (802.11ax) Bluetooth v5.4 WiFi 7 (802.11be)	802.11b 802.11g 802.11a WiFi 4 (802.11n) WiFi 5 (802.11ac) WiFi 6 (802.11ax) Bluetooth v5.4 WiFi 7 (802.11be)
Expansion Slots		
PCI Express 5.0 x16	2	1
PCI Express 3.0 x16	N/A	1
Internal I/O & Connectors		
Headers	4 x System Fan Header 1 x Water Pump Header 2 x CPU Fan Header	1 x CPU Fan Header 1 x Water Pump Header 2 x System Fan Header
USB	3 x USB 2.0 Headers (6 Ports) 2 x USB 3.2 Gen 1 Type-A Headers (4 Ports) 2 x USB 3.2 Gen 2x2 Type-C Headers (2 Ports)	2 x USB 2.0 Headers (4 Ports) 1 x USB 3.2 Gen 1 Type-A Header (2 Ports) 1 x USB 3.2 Gen 2 Type-C Header (1 Port)
Other I/O	3 x Addressable LED Headers	3 x Addressable LED Headers
Power		
Power Connectors	1 x 24-pin ATX Connector 2 x 8-pin +12 V Connector 1 x 8-pin PCIe Power Connector	1 x 24-pin ATX Connector 1 x 8-pin +12 V Connector
ATX Connector	24-pin ATX Power Connector	24-pin ATX Power Connector
Physical Specifications		
Width	9.61 in. (244.00 mm)	9.61 in. (244.00 mm)
Depth	12.00 in. (304.80 mm)	9.61 in. (244.00 mm)
Box Dimensions (WxDxH)	N/A	10.60 x 2.70 x 10.60 in. (269.24 x 68.58 x 269.24 mm)
Form Factor	ATX	microATX

THANK TOU